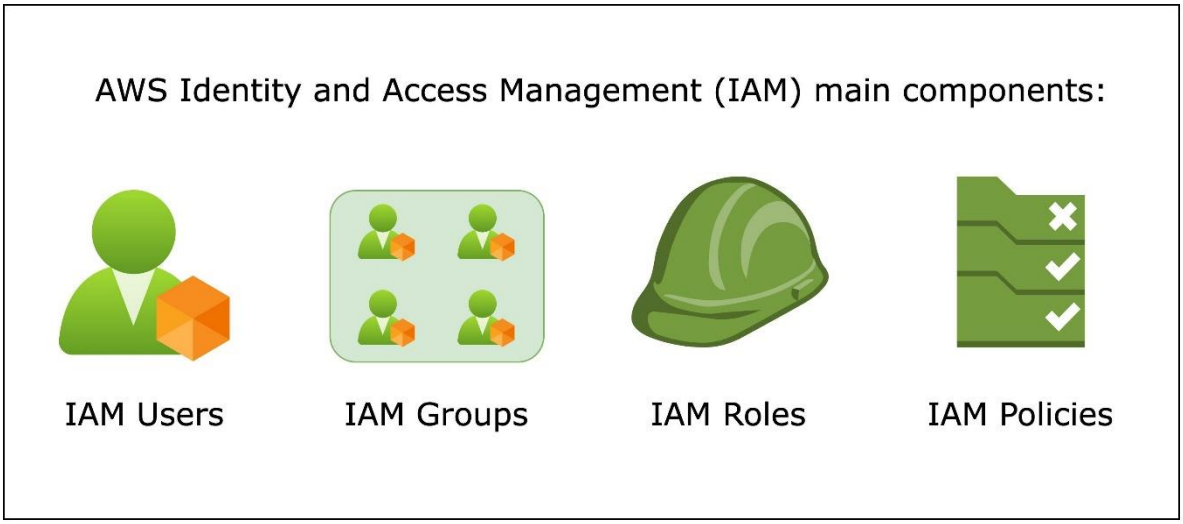
AWS

Date:29-10-2026

1.IAM Users, Roles, Policies, and Trust Relationships:

What is AWS IAM:

AWS IAM (Identity and Access Management) is a service that helps you control who can access your AWS resources and what actions they can perform.



1. IAM User

Represents an identity that needs access to AWS, often an individual person or an application requiring long-term credentials.

Example: A developer who needs to access AWS services.

2. IAM Role

Provides temporary permissions that a trusted user, AWS service, or other identity can assume.

Example: An EC2 instance assumes a role to read files from S3 without storing access keys on the instance.

3. IAM Policy

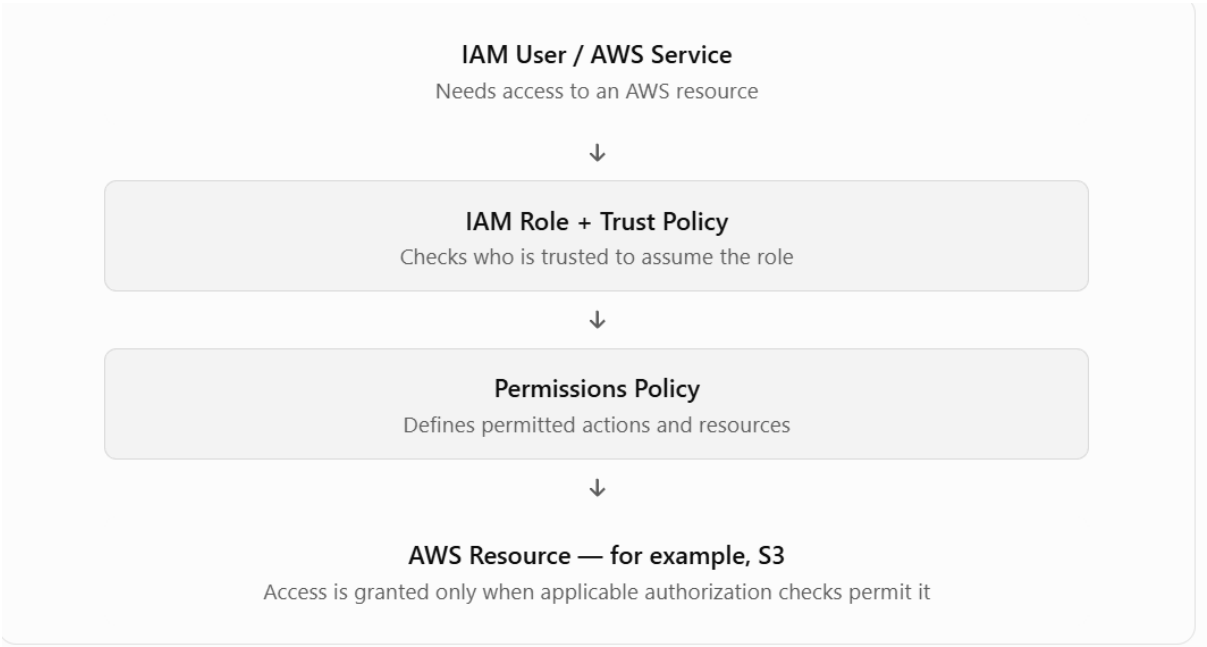
A JSON document that defines which actions are allowed or denied on which resources, optionally under specific conditions.

Example: Allow a user to list objects in a particular S3 bucket.

4. Trust Relationship

Defines who or what is allowed to assume an IAM role. It is specified in the role's trust policy.

Example: Allow the EC2 service to assume a particular IAM role.



2.Least-Privilege Access and Permission Management in AWS:

What is least privilege:

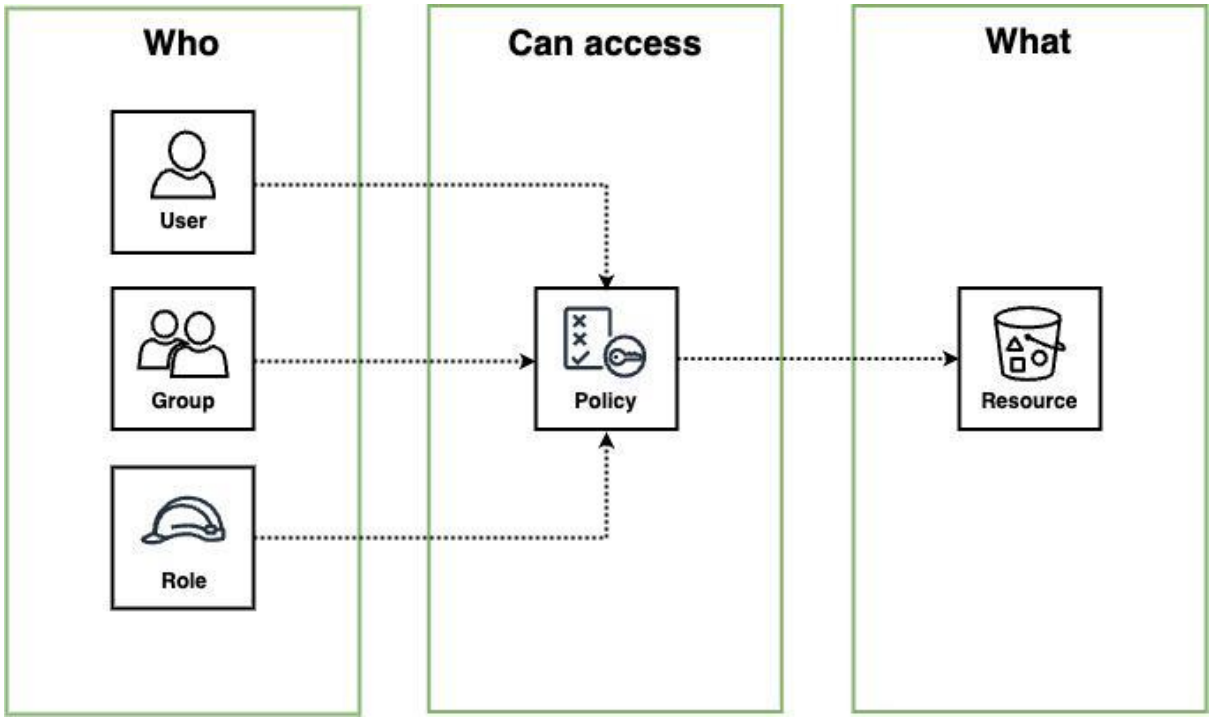
The principle of least privilege means giving a user, application, or AWS service the minimum permissions required to perform its task.

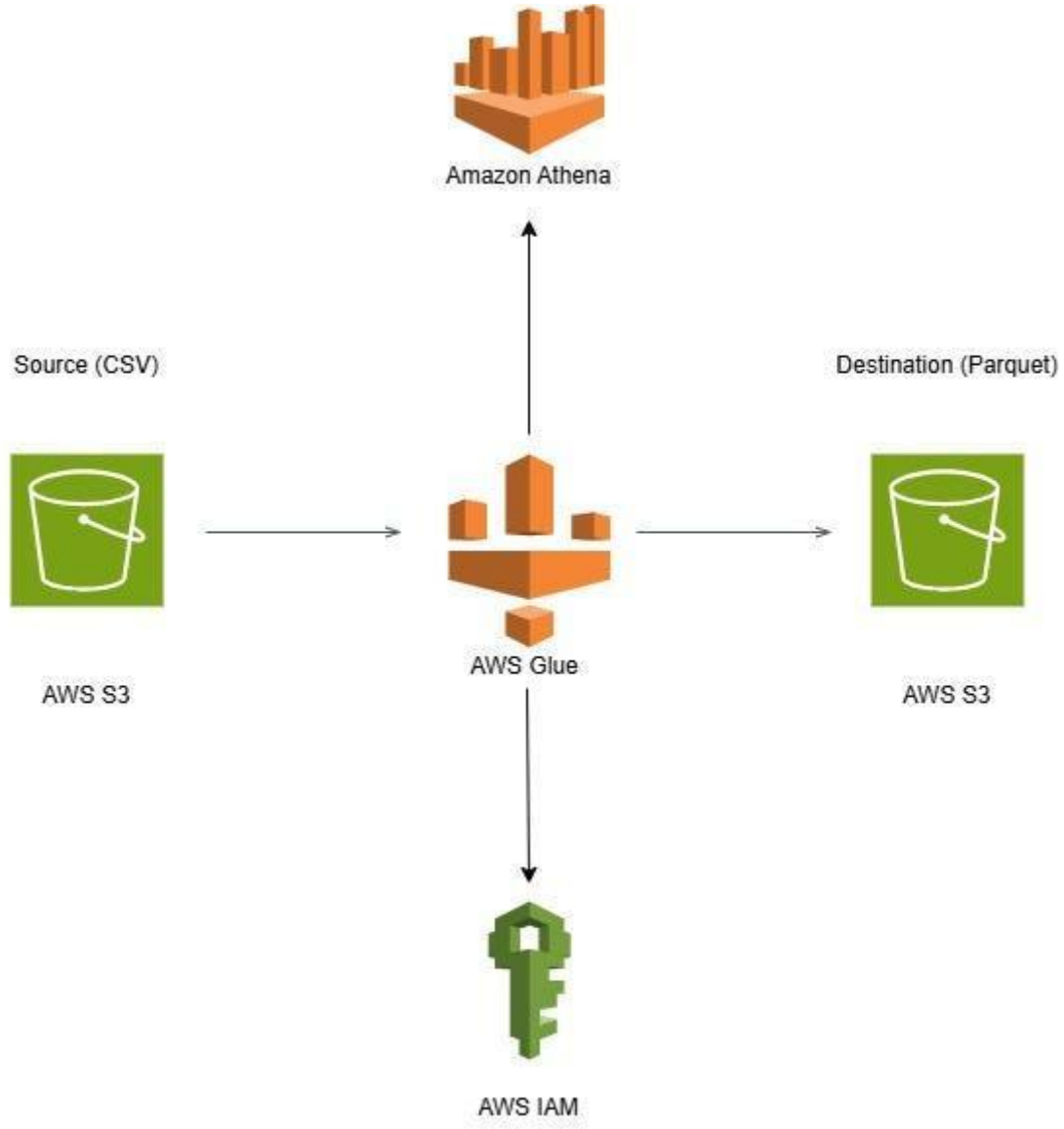
Understand permission management

AWS IAM permissions are primarily managed using policies.

Permission	AWS action example	Purpose
List bucket	`s3:ListBucket`	View objects within a bucket
Read object	`s3:GetObject`	Download or read an object
Upload object	`s3:PutObject`	Upload a file

Delete object	`s3:DeleteObject`	Delete a file





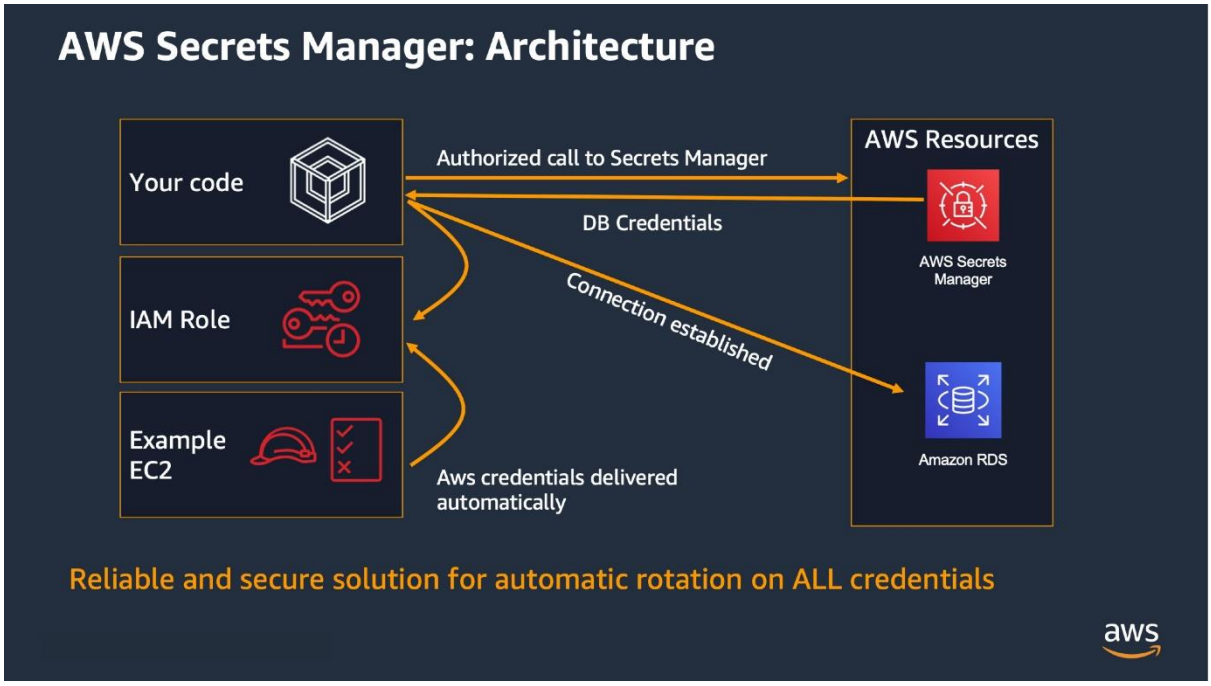
3.AWS Secrets Manager and AWS KMS:

1. What is AWS Secrets Manager?

AWS Secrets Manager is a service used to securely store, retrieve, and manage sensitive information called secrets.

Examples of secrets:

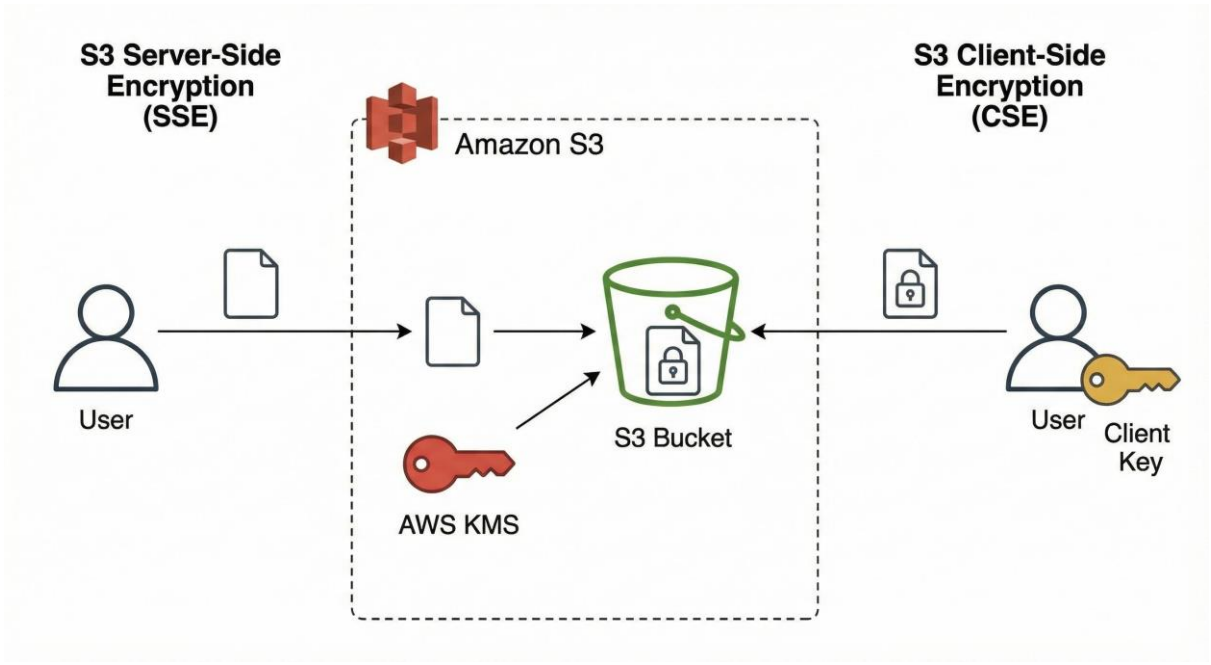
- Database usernames and passwords.
- API keys.
- Application credentials.
- Authentication tokens.



2. What is AWS KMS?

AWS KMS stands for AWS Key Management Service.

It is used to create and manage cryptographic keys that protect data through encryption and decryption.



Important KMS concepts:

Concept	Meaning
KMS Key	Cryptographic key used for encryption operations
Encryption	Converts readable data into unreadable ciphertext
Decryption	Converts ciphertext back into readable data
Customer managed key	Key that you create and manage in KMS
AWS managed key	Key managed by AWS for a particular service

3. Secrets Manager vs KMS:

Feature	Secrets Manager	AWS KMS
Main purpose	Store sensitive information	Manage encryption keys
Stores passwords	Yes	Not intended for this
Stores API keys	Yes	Not intended for this
Encryption	Uses encryption, commonly through KMS	Provides cryptographic operations
Automatic rotation	Supports secret rotation	Supports automatic key rotation for eligible keys

3.Amazon S3 Buckets, Prefixes, and Object Storage:

1. What is Amazon S3?

Amazon S3 (Simple Storage Service) is an AWS object storage service used to store and retrieve files and data over the internet.

You can store:

- CSV files
- JSON files
- Excel files
- Images and videos
- Log files
- Parquet files
- Backup files



What is an S3 Bucket?

An S3 bucket is a container used to store objects (files).

What are S3 Prefixes?

A prefix is the beginning portion of an object's key that helps organize objects into logical groups.

S3 vs Traditional File Storage

Feature	Amazon S3	Traditional file storage
Storage type	Object storage	File storage
Organization	Bucket and object keys	Folders and files
Scalability	Highly scalable	Depends on storage system
Common usage	Data lakes, backups	Shared file systems
Access	API and HTTP	File system protocols

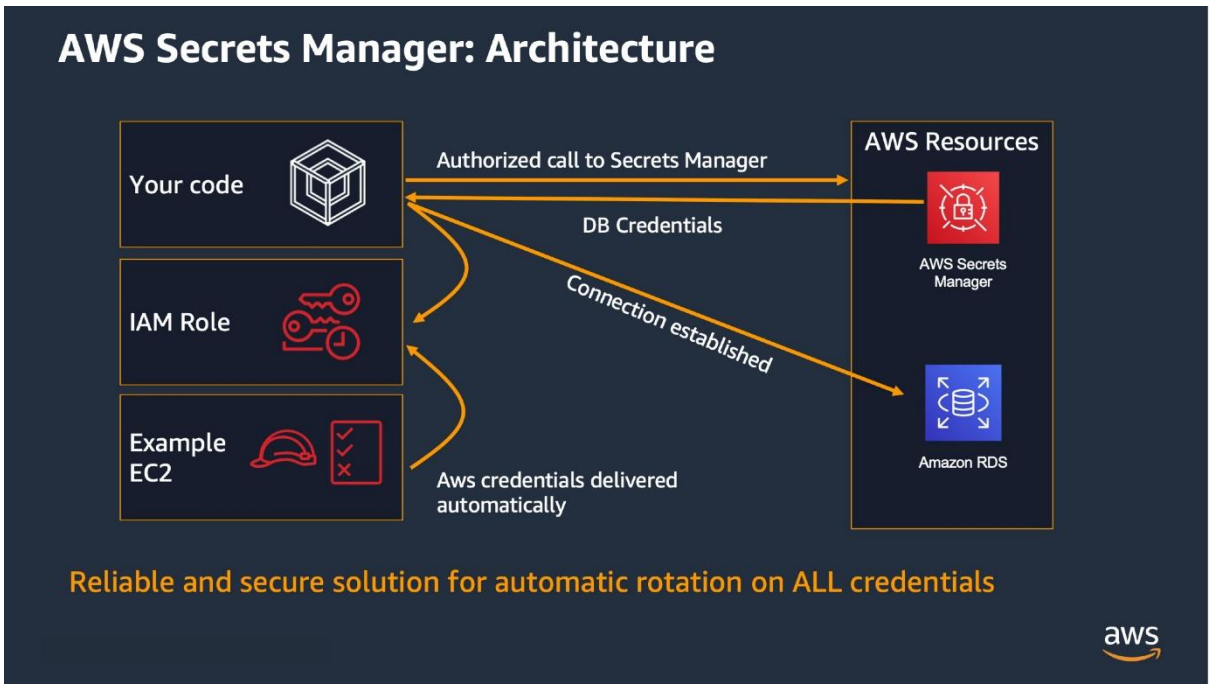
4.Secret Management and Key Rotation in AWS:

1. What is Secret Management?

Secret Management is the process of securely storing, accessing, controlling, and managing sensitive information.

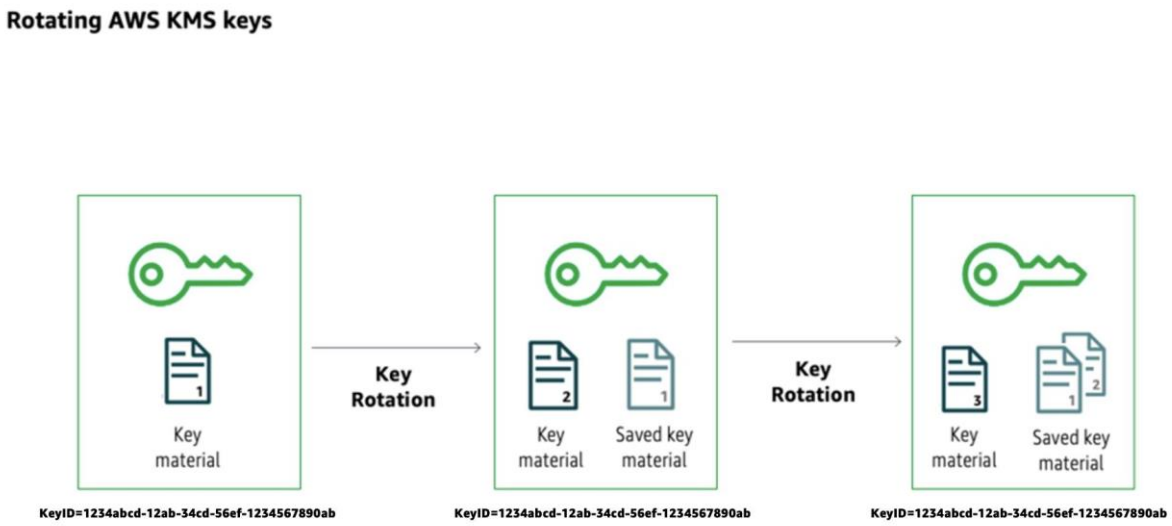
Examples of secrets:

- Database usernames and passwords.
- API keys.
- Application credentials.
- Access tokens.
- Authentication tokens.

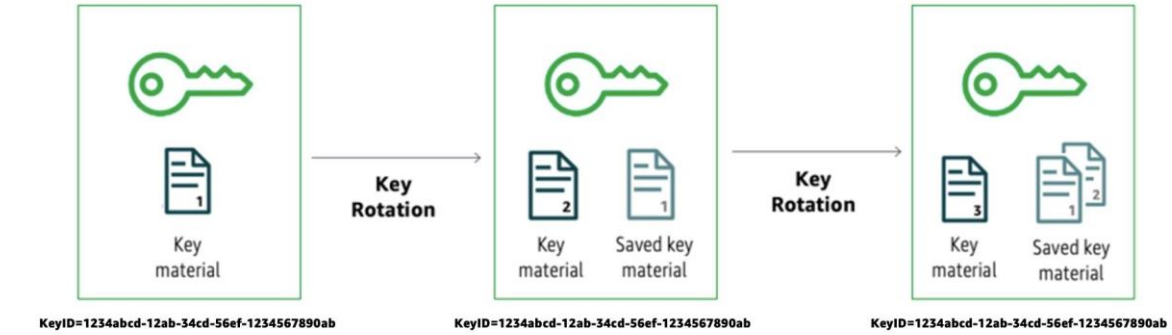


2. What is Key Rotation?

Key Rotation means periodically changing or replacing cryptographic keys to improve security.

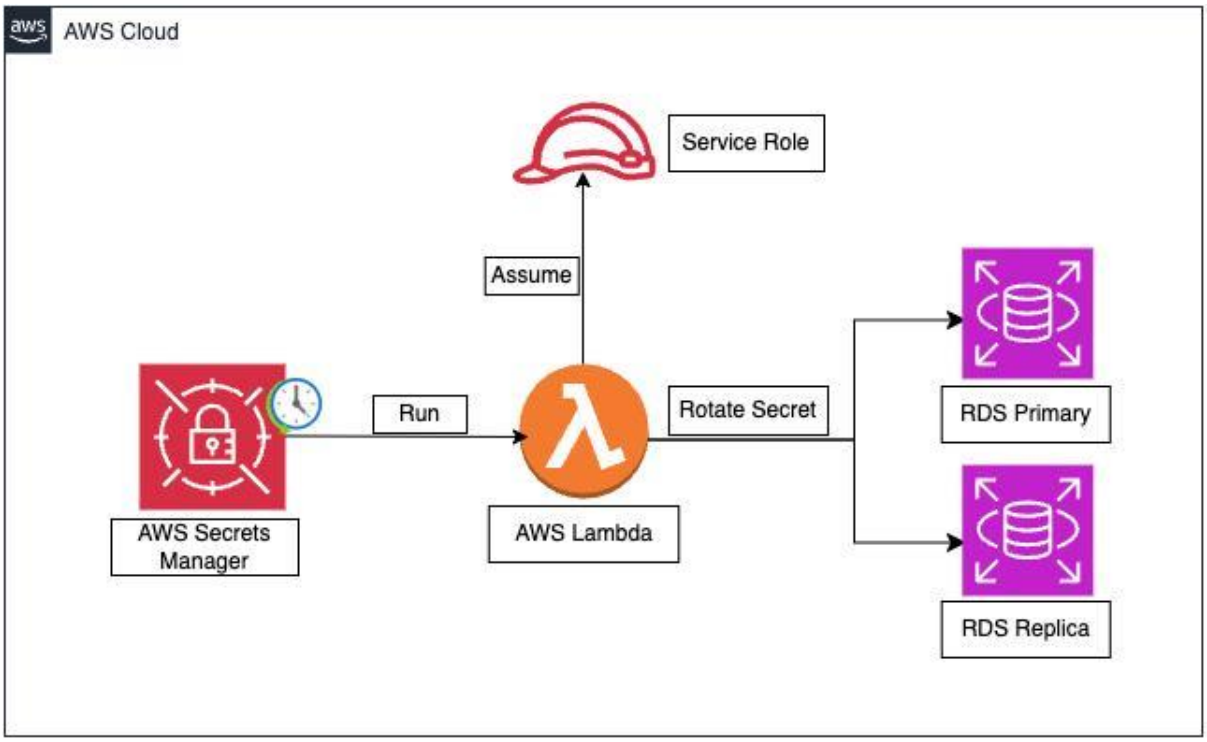


Rotating AWS KMS keys



3. What is Secret Rotation?

Secret rotation means periodically changing sensitive credentials, such as database passwords or API keys.



4. Secret Rotation vs Key Rotation

Feature	Secret Rotation	Key Rotation
Main purpose	Change credentials	Change cryptographic key material
AWS service	Secrets Manager	AWS KMS
Example	Database password	Encryption key
Frequency	Based on security requirements	Based on key policy and requirements
Automation	Rotation Lambda	KMS automatic rotation for supported keys